ISLAMIC UNIVERSITY OF TECHNOLOGY

Course Outline and Course Plan

Course Teacher	Md. Bakhtiar Hasan	**Designation**	Assistant Professor
Department	CSE	**Program**	B.Sc. Engg. CSE
Course Code	CSE 4711	**Course Title**	Artificial Intelligence
Academic Year	2025–2026	**Semester**	Winter
Course Type	Theory	**Requirement**	Compulsory
Contact Hours	3.00	**Credit Hours**	3.00
Recommended Texts	1. S. Russell, P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Prentice Hall, 2020 2. R. S. Sutton, A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018 3. D. L. Poole, A. K. Mackworth, *Artificial Intelligence: Foundations of Computational Agents*, 3rd ed. Cambridge University Press, 2023		
Prerequisites	A firm grasp of Data Structures, Discrete Mathematics, Algorithms, Probability		
Course Homepage	Google Classroom Code:		
Teaching Methods/ Approaches	☑ Lecture ☑ Demonstration ☐ Project	☐ Group Discussion ☑ Problem Solving ☐ Other:	
Teaching Aids	☑ Multimedia ☑ OHP	☑ Board and Marker ☐ Other:	

Course Details

Course Contents (As approved by the Academic Council)

Survey of concepts in artificial intelligence. Knowledge representation, search and control techniques. All machines and features of the LISP and PROLOG languages. Problem representation: search, inference and learning in intelligent systems; systems for general problems solving, game playing, expert consultation, concept formation and natural language procession: recognition, understanding and translation. Case Study on Expert Systems

Course Objectives

- Introduce the foundational principles, theoretical frameworks, and major approaches used to develop intelligent computational systems.
- Develop students' understanding of how AI problems can be represented using formal models and solved through systematic computational reasoning.
- Provide students with the mathematical and algorithmic foundations necessary to understand decision-making, learning, inference, and reasoning under uncertainty.
- Develop students' ability to compare AI techniques, understand their underlying assumptions, and analyze their effectiveness, computational characteristics, and limitations.

- Connect theoretical AI concepts to representative computational problems and practical scenarios, encouraging reasoned selection of appropriate approaches rather than uncritical reliance on established algorithms.
- - Foster a critical understanding of rational decision-making, uncertainty, and the assumptions underlying AI models and intelligent-agent behavior.

Course Outcomes (COs)

COs	CO Statements	Corresponding POs	Learning Domain and Taxonomy Levels
CO1	Apply established AI techniques, algorithms, and mathematical principles to solve well-defined computational problems and verify the resulting solutions.	PO1	C3
CO2	Formulate computational problem scenarios using appropriate formal representations and mathematical models, and justify the validity of the representations and underlying assumptions.	PO2	C6
CO3	Evaluate alternative AI techniques and recommend suitable approaches to computational problems, justifying the choices in terms of assumptions, expected behavior, effectiveness, and limitations.	PO3	C5

Mapping Table of Knowledge Profile, Complex Engineering Problem Solving, and Complex Engineering Activities

K1	K2	K3	K4	K5	K6	K7	K8	P1	P2	P3	P4	P5	P6	P7	A1	A2	A3	A4	A5
	✓		✓					✓		✓									

Lecture Plan

Weekly Plan for Course Content	
Week(s)	**Topics**
1	**Introduction:** Artificial Intelligence; Definitions and Approaches to Artificial Intelligence; Human-Like and Rational Approaches to AI; Rational Decision-Making; Utility Theory and Expected Utility Maximization; Intelligent Agents and Rational Agents; Agent–Environment Interaction; Percepts and Percept Sequences; Action Spaces and Actuators; Sensors and Perception; Rational Agent Design; Computational Models of Intelligence; Knowledge Representation and Reasoning; Search and Planning; Constraint Satisfaction Problems; Adversarial Search and Game Playing; Decision-Making under Uncertainty; Bayesian Networks; Decision Theory; Machine Learning; Natural Language Processing; Computer Vision; Robotics and Autonomous Systems; Historical Development of Artificial Intelligence; Symbolic AI and Logic-Based Reasoning; Knowledge-Based Systems and Expert Systems; Statistical Approaches to Artificial Intelligence; Neural Networks and Deep Learning; Big Data and Computational Resources in AI; Applications and Limitations of Artificial Intelligence; AI in Economics, Politics, Law, Labor, Science, and Education; Ethics and Academic Integrity in AI; Introduction to AI Problem Formulation and Problem-Solving Techniques.
2	**Uninformed Search:** Uninformed Search; Intelligent Agents and Rational Decision-Making; Reflex Agents and Planning Agents; Goal-Based Planning and Action Consequence Modeling; Search Problem Formulation; State Spaces and State-Space Abstraction; Initial States, Goal Tests, Successor Functions, and Action Costs; Search Problem Models and Environmental Representations; World States versus Search States; State-Space Size and Complexity; State-Space Graphs and Search Trees; Tree Search Algorithms; General Tree Search and Fringe

Weekly Plan for Course Content	
Week(s)	**Topics**
	Management; Node Expansion and Exploration Strategies; Depth-First Search; Breadth-First Search; Iterative Deepening Search; Uniform-Cost Search; Search Algorithm Completeness, Optimality, Time Complexity, and Space Complexity; Branching Factor, Search Depth, and Exponential Search Complexity; Cost-Sensitive Search and Path-Cost Optimization; Search Algorithm Comparison and Selection; Repeated States, Cycles, and Redundant Search; Planning versus Replanning; Model-Based Planning and Predictive Simulation; Limitations of Search and Environmental Models; Classical Search Problem Formulation and Route-Finding Applications.
3	**Informed Search:** Informed Search; Heuristic Search and Heuristic Function Design; State-Space Graphs and Weighted Search Problems; General Search Tree Algorithms and Fringe Management; Greedy Best-First Search; Uniform-Cost Search and A* Search; Path-Cost Evaluation and Heuristic Cost Estimation; Evaluation Functions and the f(n)=g(n)+h(n) Framework; Admissible Heuristics and Optimistic Cost Estimates; Consistent Heuristics and Monotonicity of Evaluation Functions; Completeness and Optimality of Informed Search Algorithms; Optimality Proofs for A* Tree Search and Graph Search; Comparison of Uniform-Cost Search, Greedy Best-First Search, and A* Search; Heuristic Design Using Relaxed Problems; Heuristic Functions for the Eight-Puzzle Problem; Manhattan Distance and Misplaced-Tile Heuristics; Heuristic Dominance and the Semi-Lattice of Admissible Heuristics; Heuristic Quality and Search Efficiency Trade-Offs; Tree Search versus Graph Search; Repeated-State Detection and Closed-Set Management; Cycle Avoidance and Search Redundancy Reduction; Conditions for Optimality in A* Graph Search; Algorithm Implementation Using Search Pseudocode; Cost-Optimal Planning and Pathfinding Applications.
4	**Constraint Satisfaction Problem I:** Constraint Satisfaction Problems (CSPs); Formal Representation of Constraint Satisfaction Problems; Variables, Domains, and Constraints; Constraint-Based Problem Formulation; Identification Problems and Assignment-Based Search; Map Coloring Problems; N-Queens Problems and CSP Formulations; Cryptarithmetic Problems; Sudoku as a Constraint Satisfaction Problem; Constraint Graphs and Binary CSPs; Types of Variables and Domains in CSPs; Finite and Infinite Discrete Domains; Continuous Variables and Continuous Constraint Satisfaction Problems; Unary, Binary, and Higher-Order Constraints; Hard Constraints and Soft Constraints; Real-World Applications of CSPs; Standard Search Formulation for CSPs; Breadth-First Search and Depth-First Search for CSPs; Backtracking Search and Recursive Backtracking Algorithms; Incremental Constraint Checking; Improving Backtracking Search; Constraint Filtering and Domain Reduction; Forward Checking; Constraint Propagation; Arc Consistency; The AC-3 Algorithm for Enforcing Arc Consistency; Limitations of Arc Consistency; Variable Ordering Heuristics; Minimum Remaining Values (MRV) Heuristic; Value Ordering Heuristics; Least Constraining Value (LCV) Heuristic; Exploiting Constraint Graph Structure and Independent Subproblems; Heuristic Search Strategies for Efficient CSP Solving.
5	**Constraint Satisfaction Problem II:** Constraint Satisfaction Problems (CSPs); Node Consistency and Arc Consistency; k-Consistency and Strong k-Consistency; Path Consistency; Constraint Propagation and Consistency Enforcement; Constraint Graphs and Problem Decomposition; Independent Subproblems and Connected Components; Tree-Structured Constraint Satisfaction Problems; Tree-Structured CSP Solving Algorithms; Topological Ordering and Variable Ordering; Arc Consistency and Domain Pruning; Backtracking-Free Search in Tree-Structured CSPs; Computational Complexity of Constraint Satisfaction Problems; Nearly Tree-Structured CSPs; Cutset Conditioning and Conditioning-Based Search; Iterative Improvement Algorithms for CSPs; Local Search for Constraint Satisfaction Problems; Complete-State Representations and Variable Reassignment; Min-Conflicts Heuristic; The N-Queens Problem; Performance Analysis of Min-Conflicts Search; Local Search and Optimization; Hill-Climbing Search; Local Maxima, Plateaus, and Ridges in Search Spaces; Random-Restart Hill Climbing; Simulated Annealing; Probabilistic Acceptance of Downhill Moves; Temperature Schedules and Convergence Guarantees; Genetic Algorithms; Fitness Functions and Population-Based Search; Selection, Crossover, and Mutation Operators; Applications and Limitations of Genetic Algorithms.

Weekly Plan for Course Content	
Week(s)	**Topics**
6	**Adversarial Search:** Adversarial Search and Game Playing; State-of-the-Art in Computer Game Playing; Formalization of Deterministic Games; Game States, Actions, Transition Functions, Terminal Tests, and Utility Functions; Game-Theoretic Foundations of Adversarial Decision-Making; Classification of Games by Determinism, Number of Players, Zero-Sum Structure, and Information Availability; Zero-Sum and Non-Zero-Sum Games; Strategies and Policies in Game Playing; Game Trees and State-Space Search; State-Value Functions and Utility Evaluation; Minimax Algorithm and Optimal Decision-Making against Rational Adversaries; Minimax Algorithm Implementation and Recursive State Evaluation; Computational Complexity of Minimax Search; Resource Constraints in Adversarial Search; Game Tree Pruning and Search-Space Reduction; Alpha-Beta Pruning Algorithm; Alpha-Beta Pruning Implementation and Search Bounds; Move Ordering and the Efficiency of Alpha-Beta Pruning; Properties and Correctness of Alpha-Beta Pruning; Metareasoning and Computational Resource Allocation; Depth-Limited Minimax Search; Evaluation Functions for Non-Terminal Game States; Feature-Based and Weighted Linear Evaluation Functions; Iterative Deepening and Anytime Search Algorithms; Search Horizon Effects and Replanning Pathologies; Evaluation Function Design for Chess and Pacman; Search Depth and the Trade-Off between Evaluation Quality and Computational Complexity; Applications of Adversarial Search to Checkers, Chess, Go, Tic-Tac-Toe, and Pacman.
7	**Uncertainty and Utility:** Probability Theory and Probability Distributions; Random Variables and Events; Laws of Probability; Conditional Probability and Evidence-Based Probability Updating; Expected Value and Mathematical Expectation; Worst-Case and Average-Case Decision-Making; Minimax and Expectimax Search Algorithms; Chance Nodes and Expected Utility Computation; Expectimax Search Pseudocode and Recursive Evaluation; Expectimax Pruning and Its Limitations; Depth-Limited Expectimax Search; Probabilistic Modeling of Opponents and Environmental Uncertainty; Informed Probabilistic Models and Opponent Modeling; Modeling Assumptions and Their Impact on Decision-Making; Optimism, Pessimism, and Risk in Adversarial Search; Search Strategy Evaluation under Model–Reality Mismatch; Mixed Adversarial and Stochastic Environments; Expectiminimax Search; Stochastic Game Playing and Backgammon; Multi-Agent Decision-Making and Utility Optimization; Non-Zero-Sum Games and Multi-Agent Utility Functions; Utility Theory and Preference Representation; Maximum Expected Utility (MEU) Principle; Rational Decision-Making under Uncertainty; Utility Functions and Agent Preferences; Axioms of Rational Preferences; Transitivity, Orderability, Continuity, Substitutability, and Monotonicity; Expected Utility Theory and the von Neumann–Morgenstern Representation Theorem; Utility Scales and Utility Normalization; Ordinal and Cardinal Utility; Invariance of Expected Utility Maximization under Positive Affine Transformations; Human Utility Elicitation and Preference Assessment; Standard Lotteries and Utility Measurement; Monetary Utility and Expected Monetary Value; Risk Aversion and Risk-Seeking Behavior; The Allais Paradox and Violations of Expected Utility Theory; Rationality, Consistent Preferences, and Utility Maximization.
8	**Markov Decision Process I:** Non-Deterministic Search; Stochastic State Transitions and Action Outcomes; Grid World Environments; Markov Decision Processes (MDPs); State Spaces, Action Spaces, Transition Models, and Reward Functions; The Markov Property and State-Based Dynamics; Policies and Optimal Policies; Expected Utility Maximization; Expectimax Search for Stochastic Decision-Making; Sequential Rewards and Utility of Reward Sequences; Discounted Cumulative Rewards and Discount Factors; Stationary Preferences and Exponential Discounting; Infinite-Horizon Decision Processes; Finite-Horizon and Infinite-Horizon Planning; Absorbing and Terminal States; Optimal State-Value Functions; Optimal Action-Value Functions; Bellman Optimality Equation; Recursive State-Value Computation; Time-Limited Value Functions; Dynamic Programming for Markov Decision Processes; Value Iteration Algorithm; Iterative Approximation of Optimal State Values; Convergence of Value Iteration; Computational Complexity of Value Iteration; Optimal Policy Extraction from State Values; Stochastic Control and Sequential Decision-Making under Uncertainty.
9	**Markov Decision Process II:** Value Iteration and Its Computational Limitations; Policy Extraction from Optimal State Values; Optimal Action Selection Using State-Value Functions;

Weekly Plan for Course Content	
Week(s)	**Topics**
	Optimal Action Selection Using Action-Value Functions (Q-Values); Policy Evaluation; State-Value Functions under Fixed Policies; Bellman Expectation Equation; Iterative Policy Evaluation; Linear System Formulation for Policy Evaluation; Policy Improvement; Policy Iteration Algorithm; Convergence and Optimality of Policy Iteration; Comparison of Value Iteration and Policy Iteration; Dynamic Programming for Markov Decision Processes; Offline Planning in Known Markov Decision Processes; Online Decision-Making under Uncertainty; Model-Based Planning versus Model-Free Learning; Introduction to Reinforcement Learning; Exploration and Exploitation in Reinforcement Learning; Sampling and Stochastic Reward Estimation; Regret and the Cost of Learning; Decision-Making under Unknown Transition and Reward Models; Fundamentals of Sequential Decision-Making under Uncertainty.
10	**Reinforcement Learning I:** Reinforcement Learning Fundamentals; Markov Decision Processes with Unknown Transition and Reward Models; Offline Planning versus Online Learning; Model-Based Reinforcement Learning; Empirical Estimation of Transition Probabilities and Reward Functions; Model-Free Reinforcement Learning; Passive and Active Reinforcement Learning; Policy Evaluation under Unknown Environment Dynamics; Direct Evaluation and Monte Carlo Value Estimation; Sample-Based Policy Evaluation; Temporal-Difference Learning; Incremental State-Value Estimation; Exponential Moving Averages and Learning-Rate Schedules; Bellman Equations and Sample-Based Approximation; Limitations of Model-Free State-Value Learning; Action-Value Functions and Q-Values; Q-Value Iteration; Q-Learning Algorithm; Sample-Based Q-Value Updates; Off-Policy Reinforcement Learning; Exploration–Exploitation Trade-Off; Learning from Stochastic Experience and Observed Rewards; Convergence of Q-Learning; Conditions for Reinforcement Learning Convergence; Optimal Policy Learning under Unknown Environment Dynamics.
11	**Reinforcement Learning II:** Reinforcement Learning; Exploration–Exploitation Trade-off; Exploration Strategies and ϵ-Greedy Action Selection; Exploration Functions and Optimistic Value Estimates; Regret Minimization in Reinforcement Learning; Model-Free Reinforcement Learning; Q-Learning and Optimal Action-Value Estimation; Generalization Across States and State-Space Abstraction; Feature-Based State and State–Action Representations; Linear Function Approximation for Value and Q-Functions; Approximate Q-Learning and Parameter Updates; Temporal-Difference Error and Bootstrapped Value Estimation; Supervised Learning and Regression for Value Function Approximation; Least-Squares Optimization and Error Minimization; Gradient-Based Learning and Stochastic Gradient Updates; Generalization and Overfitting in Function Approximation; Model Capacity and Representation Complexity; Policy Search and Direct Policy Optimization; Empirical Policy Evaluation Through Sample Episodes; Limitations of Feature-Based Value Approximation; Reinforcement Learning in Large State Spaces; Pacman-Based Applications of Approximate Q-Learning; Trade-offs Between Exploration Efficiency and Cumulative Reward; Foundations of Reinforcement Learning in Markov Decision Processes.
12	**Probability:** Probability; Random Variables and Their Domains; Probability Distributions; Joint Probability Distributions; Probabilistic Models and Representation of Uncertainty; Events and Event Probabilities; Marginal Probability Distributions and Marginalization; Conditional Probabilities and Conditional Distributions; Probability Normalization; Probabilistic Inference and Belief Updating; Inference by Enumeration; Computational Complexity of Probabilistic Inference; Product Rule of Probability; Chain Rule of Probability; Bayes' Rule and Bayesian Inference; Prior, Likelihood, and Posterior Probabilities; Diagnostic Reasoning and Causal Probability; Probabilistic Reasoning under Uncertainty; Sensor Models and Noisy Observations; Bayesian Inference for State Estimation; Probabilistic Localization and Ghost Location Estimation; Applications of Probability in Artificial Intelligence.
13	**Bayes Nets - Representation:** Probabilistic Models and Representation of Uncertainty; Probabilistic Modeling Assumptions and Model Simplification; Diagnostic Reasoning and Causal Prediction; Marginal and Joint Probability Distributions; Statistical Independence of Random Variables; Factorization of Joint Probability Distributions; Conditional Independence and Conditional Probability Distributions; Conditional Independence Assumptions in Probabilis-

Weekly Plan for Course Content	
Week(s)	**Topics**
	tic Modeling; The Chain Rule and Conditional Independence; Probabilistic Graphical Models; Bayesian Networks and Their Representation; Directed Acyclic Graphs (DAGs) and Graphical Model Notation; Nodes, Edges, and Parent Variables in Bayesian Networks; Conditional Probability Tables (CPTs); Semantics of Bayesian Networks; Factorization of Joint Distributions Using Bayesian Networks; Local and Global Dependencies in Probabilistic Models; Probabilistic Model Construction and Structure Design; Bayesian Network Examples and Applications; Joint Probability Computation in Bayesian Networks; Causal Modeling and Causal Interpretation of Bayesian Networks; Correlation, Causation, and Conditional Independence; Bayesian Network Modeling of Real-World Systems; Computational and Statistical Advantages of Bayesian Networks over Full Joint Distributions.
14	**Bayes Nets - Independence:** Bayesian Networks; Joint Probability Distributions and the Chain Rule; Bayesian Network Structure and Conditional Probability Tables; Conditional Independence Assumptions in Bayesian Networks; Graphical Representation of Conditional Independence; Causal Chains and Conditional Independence; Common-Cause Structures and Conditional Independence; Common-Effect Structures and Explaining Away; D-Separation and Conditional Independence Inference; Active and Inactive Paths in Bayesian Networks; Conditional Independence Given Evidence; Graphical Analysis of Probabilistic Dependencies; D-Separation Algorithms for Identifying Conditional Independence Relationships; Structural Implications of Bayesian Network Topologies; Representational Expressiveness of Bayesian Networks; Constraints on Joint Probability Distributions Imposed by Network Structure; Probabilistic Independence and Bayesian Network Model Complexity; Trade-Offs Between Network Structure, Representational Power, and Model Complexity.

Assessment Strategy

- **Class participation:** Class participation will be assessed with the attendance and impromptu quizzes during class periods
- **Formative assessment:** Time to time formative assessment will be carried out through written quizzes and/or homework assignments
- **Summative assessment:** Two summative assessments will be carried out through the mid and semester final examinations

Distribution of Marks

Module	Percentage (%)
Attendance	10
Quizzes/Assignments/Projects (Continuous Assessment)	30
Mid Semester Examination	30
Semester Final Examination	30
Total	**100**

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